

Prospect Theory: An Analysis of Decision under Risk

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ABSTRACT

Expected utility theory had served, by the late 1970s, as both the normative and the descriptive account of choice among risky prospects. This paper reports a systematic series of hypothetical choice experiments, administered to Israeli, Swedish, and American students and faculty, in which majorities of respondents violate the theory's independence and invariance requirements in predictable, repeatable directions. Outcomes that are certain are overweighted relative to outcomes that are merely probable (the certainty effect); reversing the sign of a prospect's outcomes reverses the preference between it and its alternative (the reflection effect); and logically equivalent formulations of the same choice yield different preferences once a riskless component is isolated (the isolation effect). Kahneman and Tversky use these regularities to construct prospect theory, in which value is defined over gains and losses relative to a reference point rather than over final asset positions, the value function is concave for gains and convex for losses, and stated probabilities are transformed by a decision-weighting function that overweights small probabilities and is subadditive, subcertain, and subproportional. The theory reduces to a small number of axiomatically grounded equations and organizes an otherwise disconnected set of anomalies, including the Allais paradox, the aversion to probabilistic insurance, and framing effects in sequential gambles.

1. Introduction

By 1979, expected utility theory occupied a nearly unchallenged position in the analysis of choice under risk. Von Neumann and Morgenstern's axiomatization had given it normative authority as a model of rational choice, and Friedman and Savage's 1948 reinterpretation of the utility-of-wealth function had extended its use as a descriptive account of ordinary economic behavior, from insurance purchases to portfolio choice. PAPER Kahneman and Tversky do not dispute the normative appeal of the expected utility axioms; they dispute the empirical claim that people actually reason in accordance with them. Using a battery of hypothetical choice problems administered to several hundred respondents, they document violations of the theory's independence axiom and its invariance requirement that are systematic, directionally predictable, and large enough in magnitude that no plausible respecification of the utility function for money can absorb them.

The paper's ambition is therefore twofold. The critique in Section 2 of the original text is narrowly empirical: it shows, problem by problem, where expected utility theory fails as a positive account of choice. The construction in Section 3 is a positive theoretical exercise: it proposes prospect theory as a replacement descriptive model, built from two primitives, a value function defined on gains and losses and a probability-weighting function, combined in a deliberately simple bilinear form that preserves as much of the mathematical structure of expected utility as the data allow. The paper is consequently reviewed here as a mixed theoretical and experimental contribution: the experiments supply the constraints that the theory must satisfy, and the theory is judged by how economically it satisfies them.

2. Research Problem and Hypothesis

2.1 Research Question

The paper asks a precisely bounded question: does the preference ordering that people reveal over pairs of monetary gambles, elicited through hypothetical choice, satisfy the independence axiom and the description invariance that expected utility theory requires, and if it systematically does not, what minimal reformulation of the axioms — retaining a bilinear combination of an outcome scale and a probability-based weight — can reproduce the observed pattern of violations without sacrificing tractability.

2.2 Motivation

The stakes of the question extend well beyond expected utility theory's descriptive accuracy in a laboratory setting. Expected utility with a concave utility function for wealth was, at the time of writing, the microfoundation for the theory of insurance demand, for portfolio selection, and

increasingly for asset pricing models built on representative-agent expected utility maximization. If the axioms fail systematically and in a direction that depends on how a problem is framed rather than on the underlying probabilities and payoffs, then models built on those axioms risk mischaracterizing the comparative statics of risk-bearing behavior, not merely its point predictions. A researcher who wishes to explain why demand for insurance against small losses coexists with demand for lottery tickets, or why otherwise identical decisions are made differently depending on whether they are framed as gains or as losses relative to a reference point, cannot appeal to a single concave utility function defined over final wealth.

2.3 Existing Literature

Three literatures converge on the paper. The first is the axiomatic tradition running from von Neumann and Morgenstern through Savage, which supplies the independence axiom under scrutiny. The second is the applied literature that had already flagged specific anomalies: Allais (1953) had shown that a concave utility function cannot simultaneously rationalize typical preferences in his two paired lotteries; Ellsberg (1961) had shown a related failure for ambiguous probabilities; Edwards (1962) and Fellner (1961, 1965) had proposed replacing objective probabilities with subjective decision weights without abandoning the expectation principle; and Markowitz (1952b), in a companion paper to his portfolio theory, had already proposed that utility be defined on gains and losses rather than on final wealth and had noted risk-seeking behavior for both small-probability gains and small-probability losses. The third literature is methodological: work by Mosteller and Nogee (1951), Davidson, Suppes, and Siegel (1957), and later Lichtenstein and Slovic (1971) on preference reversals between choice and bidding tasks, which shapes the paper's insistence on eliciting choices rather than bids or utility scale points.

2.4 Research Gap

What was missing, as of the mid-1970s, was not evidence of isolated anomalies — the Allais paradox had been known for over two decades — but a single formal model, built on primitives no more numerous than those of expected utility theory, capable of generating the certainty effect, the reflection effect, and the isolation effect simultaneously as implications of the same two functions, rather than as three unrelated ad hoc adjustments to the utility function.

2.5 Contribution

PAPER The claimed contribution is prospect theory itself: a value function v defined on deviations from a reference point, concave for gains, convex for losses, and steeper for losses than for gains, combined multiplicatively with a probability-weighting function π that is subadditive, subcertain, and subproportional, inside a bilinear evaluation rule that nests expected utility as the special case $\pi(p) = p$. **CRITIQUE** The claim is well supported for the specific class of two-outcome monetary prospects examined in the paper; it is considerably less established, and the authors are candid about this, for prospects with more than two non-zero outcomes, for non-monetary attributes, and for choices in which probabilities are not explicitly stated, all of which the Discussion section flags as directions requiring further development rather than results already in hand.

3. Theoretical Framework and Mechanism

3.1 The Three Tenets of Expected Utility under Test

The paper opens its critique by isolating exactly which components of expected utility theory are being tested. A prospect $(x_1, p_1; \dots; x_n, p_n)$ pays x_i with probability p_i , with $p_1 + \dots + p_n = 1$. Three tenets jointly define the theory being tested.

$$U(x_1 p_1; \dots; x_n p_n) = p_1 u(x_1) + \dots + p_n u(x_n) \quad (\text{Expectation})$$

$$(x_1 p_1; \dots; x_n p_n) \text{ acceptable at } w \text{ iff } U(w + x_1 p_1; \dots; w + x_n p_n) > u(w) \quad (\text{Asset Integration})$$

The third tenet, standard in economic applications rather than intrinsic to the axioms, is that u is concave ($u'' < 0$), so that the certain prospect (x) is preferred to any risky prospect with expected value x . Asset integration is the tenet that does the most work in what follows: it requires that only the resulting final wealth level matter, never the path by which it was reached or the way the choice was described. Prospect theory is best understood as an attack on precisely this tenet, retaining the bilinear, expectation-like form of the first equation while abandoning both asset integration and the identification of the weighting function with probability itself.

3.2 The Certainty Effect and the Failure of Independence

EVIDENCE The paper's Allais-type demonstration (Problems 1–2, reproduced in Table 1 below) shows that 82 percent of 72 respondents chose a certain 2,400 over an 82-chance lottery worth more in expectation, yet 83 percent of the same population reversed course once both options were scaled down by a common 0.66 probability of receiving nothing. Formally, with $u(0)=0$, the first preference implies $u(2,400) > .33u(2,500) + .66u(2,400)$, i.e. $.34u(2,400) > .33u(2,500)$; the second implies the reverse inequality, $.33u(2,500) > .34u(2,400)$. No monotone transformation of a utility-of-money function can satisfy both. The mechanism the authors infer is that removing a common 0.66 chance of winning 2,400 from both prospects "produces a greater reduction in desirability when it alters the character of the prospect from a sure gain to a probable one, than when both the original and the reduced prospects are uncertain" — the certainty effect.

3.3 The Reflection Effect

EVIDENCE Reversing the sign of every outcome in Problems 3, 4, 7, and 8 reverses the majority preference in every case (Table 1). The immediate implication is that risk aversion in the domain of gains is mirrored by risk seeking in the domain of losses: subjects who take the certain 3,000 over an 80-percent chance at 4,000 will, when the same numbers are recast as losses, take an 80-percent chance of losing 4,000 over a certain loss of 3,000. This rules out variance aversion as an alternative explanation of the certainty effect, since the certain-loss option in the mirrored problem has both a higher expected value and a lower variance than the risky option, yet is the one rejected. **INFERENCE** The reflection effect is the empirical anchor for the paper's central structural claim: that certainty increases the aversiveness of losses in exactly the same way that it

increases the desirability of gains, which is only coherent if the same overweighting-of- certainty mechanism governs both domains through a single, sign-symmetric weighting function rather than through a sign-specific attitude toward variance.

3.4 Probabilistic Insurance: A Formal Refutation of Everywhere-Concave Utility

PAPER Problem 9 asks respondents indifferent between insuring and not insuring a property whether they would purchase "probabilistic insurance" at half the premium, which pays out with probability one-half in the event of loss. Eighty percent decline it. The paper proves formally that this preference is incompatible with expected utility for any concave u . Setting $u(w-x)=0$ and $u(w)=1$, indifference between the uninsured gamble and paying premium y requires $u(w-y)=1-p$. Probabilistic insurance is preferred to full insurance if and only if

$$u(w - ry) > 1 - rp \quad (3)$$

which holds whenever u is concave, for any $0 < r < 1$. Since respondents overwhelmingly reject probabilistic insurance rather than embrace it, the concavity of u cannot by itself be the mechanism generating aversion to risk; some other force must be pushing against the small-probability, moderate- stake structure of probabilistic insurance specifically. The paper's later resolution, given in Section 3.6 below, is that the probability of loss is *overweighted*, which the proof in Section 4 of the paper shows is sufficient (given subadditivity of π) to generate exactly this preference.

3.5 The Isolation Effect and Editing

EVIDENCE Problem 10 embeds the Problem 3/4 choice inside a two-stage lottery in which a 0.75 probability of receiving nothing is common to both branches. Although the reduced compound lottery is identical in final probabilities and outcomes to Problem 4, 78 percent of 141 respondents chose as if facing Problem 3 instead — they discarded the shared first stage and evaluated only the distinguishing second-stage choice. Problems 11 and 12 show the identical mechanism operating on outcomes rather than probabilities: a common bonus added to the initial endowment is not integrated into final-wealth comparisons, so that a choice between (2,000,.5;1,000,.5) and (1,500) is answered differently depending on whether the 1,000 is framed as "already yours plus a coin flip for another 1,000" or as "a bonus of 2,000 followed by a coin-flip loss." **PAPER** The authors formalize this as an editing phase, prior to evaluation, consisting of coding (outcomes are perceived as gains and losses relative to a reference point, not as final states), combination (collapsing identical outcomes), segregation (splitting a riskless component from a risky remainder), cancellation (discarding components shared across the prospects being compared), and simplification (rounding). Editing is precisely what asset integration in expected utility theory rules out by assumption, and the isolation effect is the direct empirical signature of its presence.

3.6 The Value Function

PAPER For a regular prospect (neither strictly positive nor strictly negative), the theory's central equation is

$$V(x,p; y,q) = \Pi(p)v(x) + \Pi(q)v(y) \quad (1)$$

- V the overall value assigned to the edited prospect, defined on prospects.
- v the value function, defined on outcomes x and y , which are deviations from a reference point; $v(0)=0$.
- Π the decision-weighting function mapping stated probability p onto a weight; $\Pi(0)=0$, $\Pi(1)=1$.

For strictly positive or strictly negative prospects ($p+q=1$, and $x>y>0$ or $x<y<0$), the riskless component is segregated in editing, and evaluation instead follows

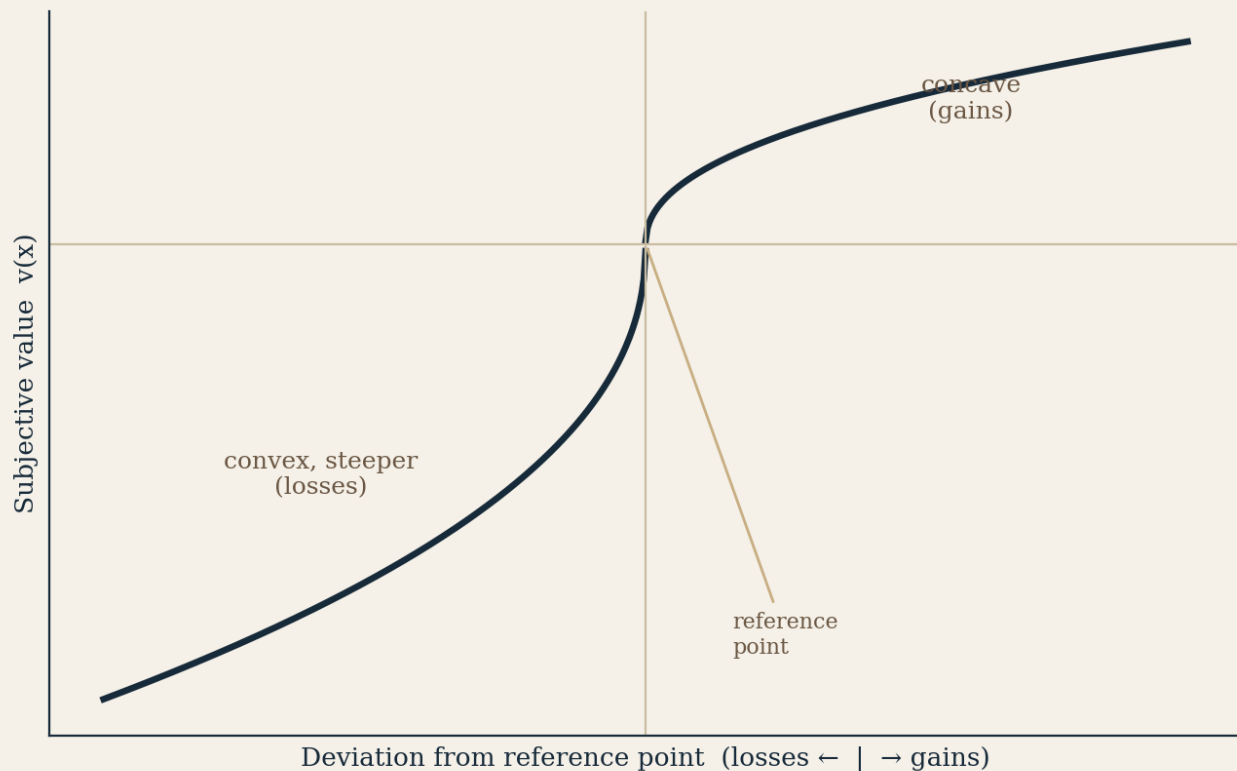
$$V(x,p; y,q) = v(y) + \Pi(p)[v(x) - v(y)] \quad (2)$$

Equation (2) collapses to equation (1) only if $\Pi(p)+\Pi(1-p)=1$, a condition the paper shows below is generally violated (subcertainty), so the two equations are genuinely distinct evaluation rules for two structurally different classes of prospect, not two notations for the same rule.

PAPER Three properties of v are established from the choice data. First, concavity for gains and convexity for losses: Problem 13 shows (6,000,.25) rejected in favor of (4,000,.25;2,000,.25), which under equation (1) implies $v(6,000) < v(4,000)+v(2,000)$; Problem 13' shows the mirrored preference for losses, implying $v(-6,000) > v(-4,000)+v(-2,000)$. Second, loss aversion — the value function is steeper for losses than for gains. From the observed aversion to symmetric bets ($y,.5;-y,.5$) that grows with stake size, the paper derives, by setting $y=0$ and letting y approach x ,

$$v(x) < -v(-x) \quad \text{and} \quad v'(x) < v'(-x) \quad (4)$$

Third, the function is defined on changes rather than final asset positions, following directly from Problems 11–12. Figure 1 below reproduces the qualitative S-shape these three properties jointly imply: steepest at the reference point, concave above it, convex and steeper below it.

Figure 1. Hypothetical value function implied by the paper's three properties

Generated for this review to illustrate the qualitative shape the paper derives (its own Figure 3); not a reproduction of the original figure and not a fitted or estimated function. Illustrative parameterization: $v(x)=x^{0.88}$ for gains, $v(x)=-2.25|x|^{0.88}$ for losses, following the curvature and loss-aversion ratio later estimated by Tversky and Kahneman (1992) for the cumulative extension of this model; the 1979 paper itself reports no fitted parameter values.

3.7 The Weighting Function

PAPER Four properties of π are derived, each from an explicit algebraic argument tied to a choice problem.

OVERWEIGHTING OF SMALL PROBABILITIES

Problem 14 shows 72 percent of 72 respondents preferring (5,000,.001) to the certain (5), which under equation (1) requires $\pi(.001)v(5,000) > v(5)$, hence $\pi(.001) > v(5)/v(5,000) > .001$ given concavity of v for gains. The mirrored insurance-purchase preference in Problem 14' implies the same conclusion from the loss side.

SUBADDITIVITY FOR SMALL P

Problem 8's preference for (6,000,.001) over (3,000,.002) implies, using $v(3,000)/v(6,000) < 1/2$ under concavity,

$$\pi(.001)/\pi(.002) > v(3,000)/v(6,000) > 1/2 \quad (5)$$

so that $\pi(rp) > r \cdot \pi(p)$ for small p and $0 < r < 1$: halving a very small probability does less than halve its decision weight.

SUBCERTAINTY

Applying equation (1) to the Allais-type reversal in Problems 1–2 (Section 3.2 above) and combining the two resulting inequalities algebraically yields

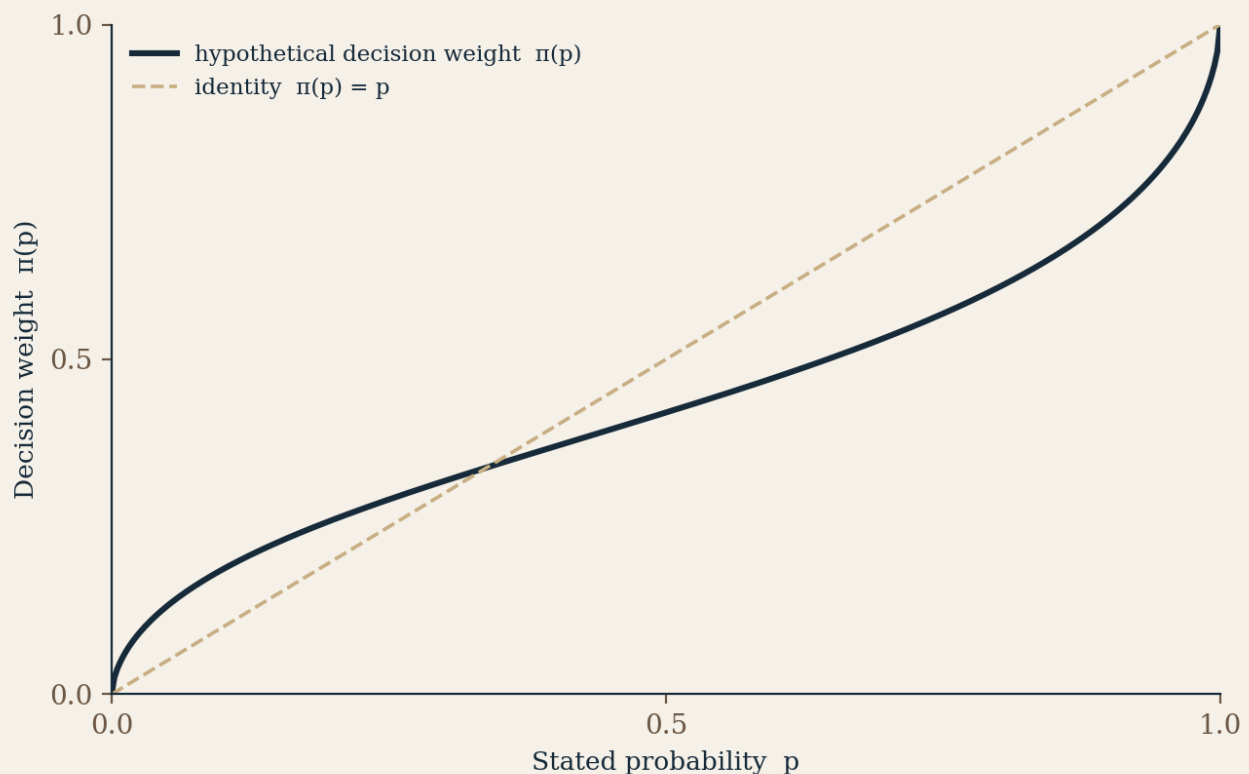
$$\Pi(.66) + \Pi(.34) < 1 \quad (6)$$

generalized as $\pi(p) + \pi(1-p) < 1$ for all $0 < p < 1$: the weights assigned to complementary uncertain events sum to less than the weight assigned to the certain event, so π is "regressive" relative to the 45° line implied by expected utility.

SUBPROPORTIONALITY

Comparing Problems 3/4 against 7/8, the same reasoning that produced (5) generalizes to: for a fixed ratio of probabilities, the ratio of decision weights $\pi(pq)/\pi(p)$ is closer to unity when probabilities are low than when they are high — a property that holds if and only if $\log \pi$ is a convex function of $\log p$, and which, combined with overweighting, is shown to imply subadditivity across the entire low-probability range.

Figure 2. Hypothetical decision-weighting function implied by subadditivity, subcertainty, and overweighting of small probabilities



Generated for this review using a one-parameter Prelec-type functional form chosen only to display the qualitative properties described in the text (the paper's own Figure 4); the specific curvature parameter is illustrative and not estimated or reported by Kahneman and Tversky (1979).

3.8 Formal Resolution of Probabilistic Insurance

PAPER Returning to Section 3.4, the paper proves that overweighting of the probability of loss ($\pi(p) > p$) together with subadditivity of π is sufficient to generate the observed rejection of probabilistic insurance. Writing $f(x) = -v(-x)$ for $x \geq 0$ (concave, since v is convex for losses), indifference $\pi(p)f(x) = f(y)$ combined with the segregated evaluation rule for mixed prospects reduces the preference for regular over probabilistic insurance to the inequality $\pi(p)/2 \geq \pi(p/2)$, which is exactly subadditivity. The value function alone cannot generate this result; the weighting function is doing the work.

3.9 Reference-Point Shifts and the Behavior of the Unadapted Loser

PAPER Section 4 of the original paper (Discussion) extends the model to cases where the reference point is not the status quo. If a decision maker who has already lost 2,000 in a venture has not yet adapted to that loss, a subsequent choice between a certain gain of 1,000 and an even chance at 2,000 or 0 may be coded as $(-2,000, .5; -1,000, .5)$ rather than $(1,000, .5; 2,000, .5)$. The paper proves, using subcertainty, that if a prospect $(x, p; -y, 1-p)$ is just acceptable, then the negatively translated prospect $(x-z, p; -y-z, 1-p)$ is strictly preferred to the certain loss $(-z)$ for $0 < z < x$. **INFERENCE** An investor sitting on an unrealized loss who has not mentally reset her reference point to current wealth is, by this mechanism, predicted to accept gambles she would reject were she to first "take the loss" and recompute a fresh reference point — a channel through which the paper's psychology maps directly onto disposition-effect-type trading behavior, though the paper itself stops at the individual decision-theoretic result and draws no market-level implication.

3.10 Axiomatic Sketch

PAPER The Appendix outlines, rather than fully proves, an axiomatization of equation (1) using additive conjoint measurement (Debreu 1960; Krantz, Luce, Suppes, and Tversky 1971). Independence, cancellation, solvability, and an Archimedean axiom yield an additive representation $V(x, p; y, q) = f(x, p) + g(y, q)$; symmetry of the prospect notation forces $f = g$. Restricting to single-outcome prospects (x, p) and imposing sign dependence yields the multiplicative form $V(x, p) = \pi(p)v(x)$. A further distributivity axiom — $(x, p; y, p) \sim (z, p)$ iff $(x, q; y, q) \sim (z, q)$ — forces the functional equation $f(\alpha) + f(\beta) = f(\gamma) \Rightarrow f(\theta\alpha) + f(\theta\beta) = f(\theta\gamma)$, whose unique monotonic solution is the power function $f(\alpha) = k\alpha^c$. Redefining scales to absorb k and c yields exactly the bilinear form of equation (1). **CRITIQUE** The sketch is explicit that it is not a complete, self-contained proof, and that the distributivity axiom — the load-bearing assumption that pins down the power-function solution and rules out other monotonic transformations — is asserted rather than independently tested against choice data anywhere in the paper.

4. Data and Empirical Strategy

4.1 Design: Hypothetical Choice Elicitation

PAPER The empirical strategy is the method of hypothetical choices: subjects are presented with pairs of prospects framed as one-shot decisions and asked which they would choose if actually facing the situation, with no real payment. Problems were assembled into questionnaire booklets of at most a dozen problems, with multiple booklet versions varying the order of problems and the left-right position of the two options in each pair, to control for order and position effects. The authors justify the hypothetical-choice method explicitly and pre-emptively rather than treating it as a limitation to be discovered later: real-choice field data cannot isolate probabilities and utilities with the precision the theory requires, and incentivized laboratory gambles are typically confined to small stakes and many repetitions of similar problems, which the authors argue introduces its own distortions and limits external validity in a different direction.

4.2 Sample

PAPER The primary sample is Israeli university students and faculty, with individual problems drawing between 64 and 141 respondents (specific N reported per problem, not pooled). Outcomes were denominated in Israeli pounds, with the paper noting that median monthly family income was approximately 3,000 Israeli pounds at the time, so that the larger stakes in the problems (around 4,000–6,000) were not trivial relative to income. A subset of problems was replicated with students and faculty at the University of Stockholm and the University of Michigan; the paper reports the resulting pattern as "essentially identical" to the Israeli results, without publishing separate percentage breakdowns for the replication samples in the body of the article.

4.3 Identification Logic

The paper's identifying strategy is a within-subject-population comparison across minimally transformed versions of the same underlying lottery: probabilities are common-ratio scaled (Problems 1 vs. 2, 3 vs. 4, 5 vs. 6, 7 vs. 8), outcomes are sign-reflected (Table I), a shared stage or shared bonus is inserted or removed (Problems 10 vs. 4, 11 vs. 12), or a lottery is decomposed into probabilistic partial coverage (Problem 9). Because each transformation is a pure change in framing or in the common component — leaving the marginal, reduced-form probabilities and outcomes identical under the reduction axiom of expected utility — any shift in majority preference across the pair is, by construction, a violation of a specific axiom (independence, invariance to representation, or reduction of compound lotteries) rather than evidence of any particular utility-of-money curvature. This is the paper's central empirical strength: it isolates which axiom fails, not merely that some numerical prediction of expected utility theory is off.

4.4 Inference

PAPER Significance is reported at the .01 level for the modal choice in each problem (denoted by an asterisk in the paper's tables), consistent with a binomial or chi-square test against a null of a 50/50 split, though the specific test statistic is not named in the text. **CRITIQUE** No correction for multiple comparisons across the fourteen numbered problems (and their sign-reversed variants) is reported or, given the era, would have been expected; with roughly twenty-five independent binary comparisons at the .01 level the paper's own reported significance levels are not the binding source of confidence in its conclusions — the replication across three subject pools and the internal consistency of the pattern (the same respondents who violate independence in Problems 1–2 also violate it, in the theoretically predicted direction, in Problems 3–4) are doing more evidentiary work than the p-values themselves.

5. Results

Table 1 reproduces the central choice problems exactly as reported, together with sample sizes and the percentage choosing each option (asterisk denotes the modal, statistically significant choice at $p < .01$ as reported by the authors).

Table 1. Selected choice problems and modal preferences (Kahneman and Tversky, 1979, Tables I–III and text)

Problem	Option A	Option B	N	% choosing A	% choosing B
1	(2,500,.33; 2,400,.66; 0,.01)	(2,400) certain	72	18	82*
2	(2,500,.33; 0,.67)	(2,400,.34; 0,.66)	72	83*	17
3	(4,000,.80)	(3,000) certain	95	20	80*
4	(4,000,.20)	(3,000,.25)	95	65*	35
3'	(−4,000,.80)	(−3,000) certain	95	92*	8
4'	(−4,000,.20)	(−3,000,.25)	95	42	58*
7	(6,000,.45)	(3,000,.90)	66	14	86*
8	(6,000,.001)	(3,000,.002)	66	73*	27
9 (insurance)	Probabilistic insurance	No probabilistic insurance	95	20	80*
10 (isolation)	Two-stage version of (4,000,.20)	Two-stage version of (3,000,.25)	141	22	78*
14	(5,000,.001)	(5) certain	72	72*	28
14'	(−5,000,.001)	(−5) certain	72	17	83*

Source: reproduced by the reviewer from the numerical values and percentages reported in Kahneman and Tversky (1979), *Econometrica* 47(2), Tables I–III and accompanying text (pp. 265–273, 281).

5.1 What the Certainty Effect Establishes

The Problem 1/2 and 3/4 pairs jointly establish that the independence axiom fails specifically when a common-ratio reduction moves one option from certain to probable, and that the effect size is large: the modal preference flips from 82–18 to 17–83 (Problems 1/2) and from 80–20 to 35–65 (Problems 3/4). **EVIDENCE** The economic magnitude is therefore not a borderline statistical artifact; it is a wholesale reversal of the modal choice, which no smooth reweighting of a fixed utility-of-money function can rationalize, since the reduction from certainty to a .99 or .80 probability event is treated as far more consequential than an equally sized reduction between two already-uncertain probabilities.

5.2 What the Reflection Effect Establishes

Every sign reversal in Table 1 flips the majority choice (Problems 3/3' and 4/4', 7/7' and 8/8', 14/14'). **EVIDENCE** This is direct evidence against any single, everywhere-concave utility-of-wealth function: concavity predicts risk aversion in both domains, but the data show risk aversion for gains and risk seeking for losses at matched probabilities and (absolute) payoffs. The economic significance is that the sign of a payoff, not only its magnitude and probability, is a first-order determinant of risk attitude — a distinction with no counterpart in a wealth-based utility function.

5.3 What Probabilistic Insurance Establishes

Eighty percent of 95 Stanford respondents reject probabilistic insurance that the paper proves is strictly preferred to full insurance under any concave expected-utility function. **EVIDENCE** Because the proof is exact rather than approximate (Section 3.4 above), this single result is sufficient, on its own, to falsify the joint hypothesis of expected utility plus concave u as a complete account of insurance demand; overweighting of the loss probability, not risk aversion over wealth levels, must be doing at least part of the work that concave utility was assumed to do.

5.4 What the Isolation Effect Establishes

The 78 percent majority in the sequential version of Problem 4 (Problem 10) directly contradicts the 65 percent majority for the mathematically identical reduced lottery in the standard version (Problem 4) — a same-population, same-final-probabilities reversal driven purely by whether a shared first stage is displayed. **EVIDENCE** This is the paper's cleanest demonstration that compound-lottery reduction, an axiom independent of and additional to independence, also fails, and that its failure is mediated by an editing operation (cancellation of shared stages) that is psychologically, not just mathematically, distinct from the certainty effect documented in Problems 1–8.

6. Robustness and Additional Tests

CONCERN: HYPOTHETICAL STAKES MAY NOT REVEAL TRUE PREFERENCES

Test — None internal to the paper; the authors instead argue methodologically that incentivized small-stakes repeated gambles have their own, arguably worse, external-validity problems. **Result** — No formal resolution offered within the paper. **Does it resolve the concern?** **CRITIQUE** Only partially. The argument that all elicitation methods have drawbacks is true but does not establish that hypothetical choice is unbiased; it establishes only that the alternative is not obviously superior. The paper's own later citation record (Section 9 below) shows that this specific concern was subsequently addressed by incentivized replications, not by anything internal to this article.

CONCERN: RESULTS MAY BE AN ARTIFACT OF ONE SUBJECT POOL OR ONE CULTURE

Test — Selected problems repeated with University of Stockholm and University of Michigan students and faculty. **Result** — "Essentially identical" pattern of preferences reported. **Does it resolve the concern?** Partially. **CRITIQUE** The replication is real and cross-national, which is a genuine strength relative to single-site experimental papers of the period, but the paper does not report the replication's own percentages, sample sizes, or significance levels in the body of the text, so the claim of "essentially identical" results cannot be independently re-assessed by a reader from the article alone.

CONCERN: ORDER AND FRAMING OF THE QUESTIONNAIRE ITSELF COULD DRIVE THE RESULTS

Test — Multiple booklet versions varying problem order and left-right position of options. **Result** — Not separately reported by version. **Does it resolve the concern?** **CRITIQUE** The randomization is a sound design choice that should attenuate simple order and position artifacts, but because breakdowns by version are not published, a reader cannot verify that the effect sizes are stable across booklet forms rather than concentrated in one.

CONCERN: INDIVIDUAL-LEVEL INCONSISTENCY VERSUS POPULATION-LEVEL MODAL REVERSAL

Test — Individual-pattern analysis reported for Problems 1–2: 61 percent of respondents made the theory-violating modal choice in *both* problems. **Result** — Majority-level consistency confirmed at the individual level for this pair. **Does it resolve the concern?** Yes, for this pair specifically. **CRITIQUE** Individual-level cross-tabulations of this kind are not reported for the other problem pairs in the paper, so the strength of within-subject consistency documented for Problems 1–2 cannot be generalized to, for example, the reflection effect pairs without further assumption.

CONCERN: IS THE VALUE FUNCTION'S SHAPE MERELY A LABORATORY ARTIFACT OF HYPOTHETICAL CHOICE?

Test — The paper cites Fishburn and Kochenberger's independent meta-analysis of thirty von Neumann–Morgenstern utility functions elicited from real business decision makers across five separate incentivized studies. **Result** — Most utility functions for gains were concave, most for losses were convex, and utility functions were considerably steeper for losses than for gains in all but one of the thirty cases. **Does it resolve the concern?** **EVIDENCE** This is the paper's strongest piece of external corroboration, because it comes from incentivized, real-decision contexts entirely

independent of the authors' own hypothetical-choice data collection, and it converges on the same S-shaped, loss-steeper functional form. It does not, however, corroborate the weighting-function properties (subadditivity, subcertainty, subproportionality), which rest entirely on the paper's own hypothetical-choice evidence.

7. Critical Evaluation

7.1 What the Paper Actually Proves

ESTABLISHED

Expected utility theory, under the joint hypotheses of the expectation principle and asset integration, is violated by a majority of respondents in every one of the paired comparisons reported; the certainty effect, reflection effect, and isolation effect are each demonstrated by an explicit, reproducible choice-problem pair with a formal proof that no monotonic transformation of a utility-of-wealth function can rationalize the observed reversal.

SUPPORTED BUT CONDITIONAL

The specific functional properties attributed to v (concave for gains, convex for losses, steeper for losses) and to π (overweighting of small p , subadditivity, subcertainty, subproportionality) are each derived from a small number of choice pairs under the maintained assumption that equation (1) or (2) is the correct evaluation rule; if respondents are instead using some other bilinear or non-bilinear rule consistent with the same reversals, the specific curvature and weighting conclusions would not follow in the same form, though the underlying reversals themselves would remain unexplained by expected utility regardless.

SUGGESTIVE

The claim that the same psychological mechanism (overweighting of certainty, editing of shared components) organizes all three effects simultaneously is a parsimony argument, not a separately tested restriction; the paper does not present a problem in which the certainty effect and the isolation effect are pitted against each other in a way that would discriminate a unified mechanism from three independent ones.

NOT ESTABLISHED

The paper explicitly does not establish a complete axiomatization of prospect theory (the Appendix is a sketch), a theory of multi-outcome (more than two non-zero outcomes) prospects, a theory of choice under unstated or ambiguous probabilities beyond a qualitative conjecture, or any market-level or general-equilibrium implication of the individual-level anomalies it documents; asset prices, insurance markets, and gambling markets are not modeled anywhere in the paper.

7.2 Strengths

Strengths recognized by the authors. The paper is explicit about the fragility of hypothetical-choice methodology (Section 2 of the original text) and about the incompleteness of its own axiomatization (Appendix) and its silence on multi-outcome prospects and non-stated probabilities (Discussion). **Strengths identified independently.** Beyond what the authors claim for themselves, the design's use of matched, minimally transformed problem pairs is unusually disciplined for 1979: each pair isolates a single axiom, so that a reversal is diagnostic of a specific failure rather than a diffuse "expected utility doesn't fit" conclusion. The theory that follows is also remarkably economical: two functions, four qualitative properties each, and a single bilinear combination rule reproduce every documented anomaly, which is a much stronger form of parsimony than the piecemeal utility-function patches (e.g., Friedman-Savage's doubly inflected u) that preceded it.

7.3 Weaknesses and Independent Critique

HYPOTHETICAL STAKES

Problem — No real money changes hands. **Why it matters** — Expected utility theory's own domain of application is real economic choice; a demonstration that people answer hypothetical questionnaires inconsistently with the theory does not, by itself, establish that they would choose inconsistently when their own wealth is at stake. **Potential bias** — Respondents may give answers that feel intuitively "right" without the deliberative effort that a large real stake would induce, potentially inflating the size of framing-driven reversals. **How it could be tested** — Repeat the key problem pairs (Problems 1/2, 3/4, 9) with real, if modest, incentives and compare effect sizes; this is exactly the research design later pursued by numerous incentivized replications, several of which (Section 9.3 below) confirm the qualitative pattern at reduced magnitude.

SELECTION AND CONSTRUCTION OF THE PROBLEM SET

Problem — The fourteen numbered problems were constructed by the authors specifically to elicit known or suspected violations (several are explicit variants of Allais 1953). **Why it matters** — A theory built to fit a curated set of anomaly-eliciting problems is not thereby shown to describe typical, unselected choice behavior. **Potential bias** — Overstatement of how pervasive these reversals are in ordinary decisions with less deliberately engineered structure. **How it could be tested** — Draw a random sample of naturally occurring risky choices (insurance take-up decisions, lottery ticket purchases) and test whether the same reversal patterns appear without the researcher choosing the problem structure ex ante; this is largely the empirical program later carried out under the broader banner of behavioral economics (Section 9 below), rather than something attempted in this paper.

MODEL CALIBRATION VERSUS MODEL TESTING

Problem — The 1979 paper qualitatively derives the properties of v and π from inequalities implied by choice data but never fits and reports numerical parameter values for either function. **Why it matters** — Without fitted parameters, prospect theory as presented here is a qualitative, not a quantitative, predictive model; it cannot, in this form, be used to generate a point prediction for

a new, previously unseen prospect. **Potential bias** — A qualitative model is harder to falsify than a fully parameterized one, since almost any reversal consistent with the four qualitative properties of π can be rationalized after the fact. **How it could be tested** — This gap motivated the authors' own later work: Tversky and Kahneman (1992) fit explicit parametric forms (a power value function and a probability-weighting function) to a new, larger dataset, converting the qualitative 1979 model into an estimable one.

THE SINGLE MOST IMPORTANT WEAKNESS

CRITIQUE Considered against the paper's own ambitions, the single most consequential weakness is the absence of any market-level bridge. The paper demonstrates, convincingly, that individual choice under risk violates expected utility in specific, structured ways; it says nothing about whether these individual-level anomalies survive aggregation, market discipline, learning, and arbitrage well enough to matter for asset prices, insurance premia, or gambling-market equilibria. This is not a flaw in what the paper claims to do — it is explicitly a paper about individual decision theory — but it is the gap that the subsequent thirty years of behavioral finance literature (Section 9 below) exists specifically to fill, and a reader using this paper as a foundation for asset-pricing work must treat that bridge as entirely unbuilt within the 1979 text itself.

8. Contradictions and Edge Cases

8.1 Under what conditions should the certainty effect disappear?

INFERENCE By the paper's own logic, the certainty effect should attenuate as both compared probabilities move away from the neighborhood of 1 or 0, since subcertainty describes a maximal wedge near the endpoints of the probability interval and the weighting function is described as comparatively "shallow" in the interior (Figure 2 above). Problems 7 and 8 partially confirm this: at substantial probabilities (.90, .45) subjects choose according to expected value ranking, while at minuscule probabilities (.002, .001) they choose according to the larger prize — consistent with a weighting function that is close to linear in an intermediate range and diverges from linearity mainly near the endpoints.

8.2 What happens at extreme values?

CRITIQUE The paper itself flags an internal tension at the probability endpoints: strict monotonicity of π together with subadditivity and the discontinuity implied by "a limit to how small a decision weight can be attached to an event, if it is given any weight at all" means π is explicitly described as "not well-behaved near the end-points." A function that is simultaneously required to be continuous enough to support the algebraic derivations in Sections 3.6– 3.8 above and discontinuous at 0 and 1 is a genuine internal tension that the paper acknowledges but does not resolve; it is resolved only later, by Tversky and Kahneman's (1992) cumulative version of the theory, which replaces the simple probability weighting of single outcomes with a weighting of cumulative probability ranks and removes the need for an ad hoc discontinuity.

8.3 Does the mechanism depend on a specific parameter, or could another mechanism generate the same reversals?

CRITIQUE Two entirely different accounts could, in principle, generate the Problem 1/2 reversal without any change in decision weights: regret theory (later formalized by Loomes and Sugden, 1982) explains Allais-type reversals through anticipated regret over forgone outcomes rather than through probability weighting, and rank-dependent expected utility (Quiggin, 1982) achieves a similar reweighting of probabilities but through a different, rank-ordered aggregation rule rather than prospect theory's simple pairwise weighting. **INFERENCE** Because the 1979 paper's tests are constructed to discriminate expected utility from a theory with decision weights, not to discriminate among competing weighted-probability theories, the choice data alone cannot by themselves rule out these alternative formalizations; discriminating among them requires problems in which prospect theory, regret theory, and rank-dependent utility make different numerical predictions, which is outside the scope of this paper.

8.4 Does the reflection effect survive when losses are unavoidable rather than merely possible?

EXTENSION All the paper's loss-domain problems (Table I, right column) present a probabilistic loss against a certain, smaller loss, never a forced, unavoidable large loss against which no alternative exists. Whether risk-seeking for losses persists, weakens, or reverses when an outcome is catastrophic and irreversible rather than merely a moderate hypothetical sum (as in mortality risk, bankruptcy, or existential threats to a firm) is not tested here and remains an open empirical question that the paper's own scope, calibrated to sums on the order of a month's income, cannot answer.

8.5 Does the isolation effect survive outside artificially sequential framings?

CRITIQUE Problem 10's two-stage structure is explicitly constructed by the experimenter; naturally occurring sequential decisions (a two-round investment, a multi-period insurance renewal) may or may not be mentally represented by decision makers in the same "sequential-form" way the experimenter imposes. The isolation effect is therefore conditional on how a real-world compound decision is *perceived* to be structured, a perceptual step the paper does not model and cannot, by its own hypothetical-choice design, measure independently of the choice outcome it is meant to explain.

9. Further Literature

9.1 Preceding the Paper

Allais (1953) — Shared idea: a paired-lottery demonstration that the independence axiom fails. Relationship: Problems 1–2 are an explicit, moderate-stakes variant of Allais's original large-stakes example. Why it matters: establishes that the anomaly predates prospect theory by a quarter-century; Kahneman and Tversky's contribution is the general theory that organizes it, not the discovery of the anomaly itself. **Friedman and Savage (1948)** — Shared idea: a utility-of-wealth

function reconciling simultaneous insurance-buying and gambling. Relationship: represents the expected-utility-preserving alternative that prospect theory displaces; Friedman-Savage keep asset integration and the expectation principle, using a doubly inflected $u(w)$ instead. Why it matters: shows the paper's proposed break with the expectation principle was not the only conceivable repair available at the time, and the two accounts make different predictions precisely where reference-dependence bites (Problems 11–12). **Markowitz (1952, "The Utility of Wealth")** — Shared idea: value defined on gains and losses rather than final wealth, and risk-seeking for small-probability gains. Relationship: is cited repeatedly in the 1979 paper itself as the direct intellectual predecessor of the value-function idea; Kahneman and Tversky's own Figure 3 is explicitly contrasted with Markowitz's function, which is "relatively shallow" near the reference point where theirs is steepest. Why it matters: the value function's core structural idea is not new to this paper; what is new is combining it with an explicit, separately derived weighting function and a disciplined choice-based elicitation. **Edwards (1962); Fellner (1961, 1965)** — Shared idea: replacing probabilities with subjective decision weights. Relationship: anticipates the weighting function π but retains the expectation principle in its simple form and does not derive the specific subadditivity, subcertainty, and subproportionality properties from choice data.

9.2 Contemporary and Immediately Connected Work

Tversky and Kahneman (1974), "Judgment under Uncertainty: Heuristics and Biases," Science — Shared idea: systematic, direction-predictable departures from normative statistical reasoning. Relationship: the companion paper on probability *judgment* (as opposed to probability *weighting* in choice), cited directly in the Discussion section as the source of biases that could additionally distort the perceived likelihood feeding into π . **Lichtenstein and Slovic (1971); Grether and Plott (1979)** — Shared idea: preference reversals between choice and bidding tasks. Relationship: cited directly to justify the paper's insistence that v and π be measured from choices rather than from bids or bidding-based utility elicitation. **Ellsberg (1961)** — Shared idea: departures from expected utility under ambiguous, as opposed to merely risky, probabilities. Relationship: cited in the Discussion as the source for the conjecture, left untested in this paper, that decision weights are further reduced under ambiguity.

9.3 Later Development, Extension, and Application

Tversky and Kahneman (1992), "Advances in Prospect Theory: Cumulative Representation of Uncertainty," Journal of Risk and Uncertainty — Shared idea: value and weighting functions combined to evaluate risky prospects. Relationship: directly extends this paper by replacing outcome-by-outcome weighting with cumulative-rank weighting, resolving the endpoint discontinuity flagged in Section 8.2 above and, crucially, fitting explicit parametric forms (power value function with loss-aversion coefficient near 2.25, and a Prelec-type weighting function) to new incentivized choice data spanning gains, losses, and mixed prospects with more than two outcomes. This is the paper's own direct sequel and the version of the theory most commonly used in applied work today. **Benartzi and Thaler (1995), "Myopic Loss Aversion and the Equity Premium Puzzle," Quarterly Journal of Economics** — Shared idea: loss aversion combined with a specific evaluation horizon. Relationship: applies the loss-aversion property of v derived here, together with an assumption about how frequently investors evaluate portfolio outcomes, to argue

that the historical equity premium is consistent with reasonable risk preferences once loss aversion and frequent mental "repricing" are taken into account; this is precisely the market-level bridge the 1979 paper itself does not attempt (Section 7.3 above). **De Long, Shleifer, Summers, and Waldmann (1990), "Noise Trader Risk in Financial Markets," Journal of Political Economy** — already reviewed in this library under Behavioral Finance. Shared idea: departures from a single rational, wealth-maximizing agent as the source of persistent price effects. Relationship: complementary rather than derivative — DSSW models the market consequences of irrational *beliefs* about future prices, while Kahneman and Tversky model irrational *evaluation* of a given, correctly perceived set of probabilities and outcomes; a fully specified behavioral asset-pricing model plausibly needs both channels. **Barberis, Huang, and Santos (2001), "Prospect Theory and Asset Prices," Quarterly Journal of Economics** — Shared idea: loss aversion and a reference-dependent value function embedded in a representative-agent consumption-based asset-pricing model. Relationship: is the most direct attempt to build the missing market-level bridge identified in Section 7.3, showing that loss aversion over financial wealth fluctuations, combined with prior gains and losses affecting risk aversion (the "house money" mechanism foreshadowed in Section 3.9 above), can generate a high equity premium and excess volatility inside a general-equilibrium asset-pricing model. **Rabin (2000), "Risk Aversion and Expected-Utility Theory: A Calibration Theorem," Econometrica** — Shared idea: none directly, but a foundational later critique of expected utility that independently reinforces this paper's conclusion. Relationship: proves that any expected-utility explanation of modest risk aversion over small stakes implies absurd, essentially infinite risk aversion over large stakes, formally closing off the possibility that the anomalies documented here could, after all, be rescued by some sufficiently clever concave utility-of-wealth function; the calibration theorem is the rigorous complement to the informal Section 3.4 proof given in the 1979 paper itself.

10. Research Gaps

10.1 Paper-Level Gaps

Gap — No parametric estimates of v or π . **Why it matters** — without fitted functional forms the theory cannot generate quantitative point predictions. **Existing limitation** — addressed only in the 1992 sequel, not here. **Possible way forward** — already realized historically by Tversky and Kahneman (1992); a live gap is whether the specific 1992 parameter values are stable across populations, stakes, and decades, which is an ongoing estimation question rather than a settled one.

10.2 Literature-Level Gaps

Gap — The market-level implications of individually loss-averse, probability-weighting agents remain only partially worked out relative to the depth of the individual-level evidence. **Why it matters** — asset prices are set by markets, not by isolated individuals, and arbitrage or aggregation could in principle wash out individual-level anomalies. **Existing limitation** — models such as Barberis, Huang, and Santos (2001) show existence-proof style results in stylized representative-

agent settings, not a fully general account of when and how much individual-level prospect-theoretic distortions survive aggregation across heterogeneous investors. **Possible way forward** — richer heterogeneous-agent asset-pricing models in which only a subset of investors are prospect-theoretic and the rest are expected-utility arbitrageurs, in the spirit of the limits-to-arbitrage literature already in this library (De Long, Shleifer, Summers, and Waldmann, 1990).

10.3 Methodological Gaps

Gap — Reliance on hypothetical, unincentivized choice. **Why it matters** — external validity of the specific effect sizes for high-stakes, real-money decisions remains open. **Existing limitation** — the paper offers a methodological defense but no incentivized within-paper test. **Possible way forward** — modern within-subject incentivized designs with real, randomly-paid-out stakes at magnitudes comparable to those in the original problems, ideally in a pre-registered replication across multiple subject pools and stake levels, to map out exactly how the effect sizes documented in Table 1 above scale with real financial exposure.

11. Research Opportunities

1. Incentivized, high-stakes replication of the certainty and reflection effects

RESEARCH QUESTION: Do the certainty and reflection effects documented in Problems 1–8 persist, at comparable magnitude, when real payoffs of the same size relative to income are actually paid out?

MOTIVATION: Directly addresses the paper's own acknowledged and never internally resolved hypothetical-choice limitation (Section 7.3).

HYPOTHESIS: Effect sizes attenuate but do not disappear under real incentives, consistent with the broader incentivized-replication literature summarized in Section 9.3.

ECONOMIC MECHANISM: If overweighting of certainty and loss aversion are genuine features of the value and weighting functions rather than artifacts of costless hypothetical reasoning, the reversals should survive real stakes at reduced but nonzero magnitude.

LITERATURE CONNECTION: Directly tests the boundary between Kahneman and Tversky (1979) and the incentivized-experiment literature it explicitly declines to conduct itself.

DATA: New laboratory data collected by the researcher, using the exact problem wording (scaled to local currency and income) from Problems 1, 2, 3, 4, 7, 8, and 9.

VARIABLES: Binary choice indicator; randomly selected problem paid out for real at the session's end (random-lottery incentive system).

METHOD: Within-subject repeated-measures design with a hypothetical-choice control arm and a real-incentive treatment arm, randomly assigned.

IDENTIFICATION STRATEGY: Random assignment to hypothetical versus incentivized arms isolates the effect of real stakes from any other confound.

EXPECTED CONTRIBUTION: A direct, modern estimate of how much of the 1979 effect sizes is attributable to the hypothetical-choice method itself.

MAIN RISK: Ethics-board payout ceilings may cap feasible stakes well below the income-relative magnitudes used in the original Israeli-pound problems, limiting the comparison.

2. Testing reference-point adaptation and the "unrealized loss" prediction with real portfolio data

RESEARCH QUESTION: Do investors who are sitting on unrealized brokerage-account losses take on measurably more portfolio risk in the following period than investors with unrealized gains of the same magnitude, controlling for wealth and risk tolerance?

MOTIVATION: Directly tests the Section 3.9 reference-point-shift proposition, which the 1979 paper proves theoretically but never tests against real financial behavior.

HYPOTHESIS: Unrealized-loss positions predict subsequent increases in portfolio volatility relative to matched unrealized-gain positions.

ECONOMIC MECHANISM: Failure to adapt the reference point to current wealth after a loss induces risk-seeking over the domain framed as further losses, per the formal proof in Section 3.9.

LITERATURE CONNECTION: Connects the 1979 individual-choice result to the disposition-effect literature (Shefrin and Statman, 1985; Odean, 1998) without assuming the reader already accepts prospect theory as the mechanism.

DATA: Retail brokerage account panel data with position-level cost basis, trade history, and portfolio composition.

VARIABLES: Unrealized gain/loss relative to cost basis; subsequent portfolio beta or realized volatility; account-level fixed effects.

METHOD: Panel regression with account and time fixed effects, unrealized position status as the key regressor.

IDENTIFICATION STRATEGY: Within-account, within-time variation across simultaneously held positions with different unrealized gain/loss status, controlling for account-level risk preference via fixed effects.

EXPECTED CONTRIBUTION: A direct, market-data test of a specific, previously untested theoretical proposition from the 1979 paper.

MAIN RISK: Confounding from tax-loss harvesting incentives, which independently predict trading around unrealized losses for reasons unrelated to reference-point adaptation.

3. Discriminating prospect theory from regret theory and rank-dependent utility using tailored problem triples

RESEARCH QUESTION: Can a set of choice problems be constructed in which prospect theory, regret theory, and rank-dependent expected utility make distinct, mutually exclusive numerical predictions, and which theory best fits observed choices?

MOTIVATION: Addresses the edge case raised in Section 8.3: the 1979 paper's problems cannot discriminate among competing decision-weight theories that all violate independence.

HYPOTHESIS: No single theory dominates uniformly; different theories fit best in different regions of the probability-outcome space.

ECONOMIC MECHANISM: Each theory attributes the Allais-type reversal to a different psychological process (weighting of stated probability, anticipated regret, or rank-dependent weighting of cumulative probability), which is only observationally distinguishable in carefully engineered problem structures.

LITERATURE CONNECTION: Directly engages Loomes and Sugden (1982) and Quiggin (1982) as the two principal rivals to the 1979 model identified in Section 8.3.

DATA: New incentivized experimental data using purpose-built discriminating problem triples.

VARIABLES: Choice pattern across the discriminating triple; model-implied likelihood under each candidate theory.

METHOD: Structural model comparison (e.g., Bayesian model averaging or a mixture-of-types model) fit to individual-level choice data.

IDENTIFICATION STRATEGY: Problems are constructed so that each theory's prediction is a distinct vertex of the choice space, ruling out observational equivalence by design.

EXPECTED CONTRIBUTION: A sharper answer to which specific mechanism, among several theoretically plausible ones, best organizes Allais-type reversals.

MAIN RISK: Constructing genuinely discriminating problems is technically demanding, and near-indifference in any one theory's prediction can make the test under-powered.

4. Testing loss aversion and probability weighting in algorithmic and robo-advised portfolio decisions

RESEARCH QUESTION: Do investors who delegate portfolio decisions to a rules-based robo-advisor exhibit weaker certainty-effect and reflection-effect signatures in their override and withdrawal behavior than investors who make discretionary trades?

MOTIVATION: A genuinely new empirical setting (algorithmic delegation) that did not exist in 1979 and that plausibly changes how the editing phase described in Section 3.5 operates, since the robo-advisor pre-performs much of the framing.

HYPOTHESIS: Discretionary-override and early-withdrawal decisions during market drawdowns show the reflection-effect signature (risk-seeking in the loss domain, i.e., "doubling down" or panic-selling depending on framing); passive, non-overriding accounts do not.

ECONOMIC MECHANISM: Removing the investor from the moment-to-moment editing and evaluation of gains and losses (by delegating to an algorithm) is hypothesized to suppress the reference-dependent reversals documented in Sections 3.3 and 3.9.

LITERATURE CONNECTION: Extends the 1979 value-and-weighting-function framework, together with the reference-point-adaptation result of Section 3.9, to a delegation setting not contemplated by either the original paper or the disposition-effect literature it inspired.

DATA: Robo-advisory platform account-level data on override frequency, timing relative to drawdowns, and matched discretionary-brokerage account data as a comparison group.

VARIABLES: Override/withdrawal indicator; contemporaneous unrealized gain/loss; market drawdown magnitude; account type (delegated versus discretionary).

METHOD: Difference-in-differences comparing override behavior around large drawdowns across delegated and discretionary account types.

IDENTIFICATION STRATEGY: Market-wide drawdowns provide a common shock; the delegated-versus-discretionary distinction provides the cross-sectional comparison.

EXPECTED CONTRIBUTION: Evidence on whether algorithmic delegation is a practical debiasing mechanism against the specific anomalies this paper documents.

MAIN RISK: Selection into delegated versus discretionary accounts is unlikely to be random, so investor-level risk preference must be carefully controlled for or instrumented.

5. Re-estimating the weighting function's endpoint behavior with modern non-parametric methods

RESEARCH QUESTION: What is the actual shape of the decision-weighting function in the immediate neighborhoods of $p=0$ and $p=1$, where the 1979 paper itself concedes π is "not well-behaved"?

MOTIVATION: Directly resolves the internal tension flagged in Section 8.2 between the continuity required for the paper's algebraic derivations and the discontinuity the authors themselves describe near the endpoints.

HYPOTHESIS: The weighting function exhibits a sharp but continuous (not literally discontinuous) transition near the endpoints, better captured by a flexible non-parametric estimator than by the two-parameter forms used since Tversky and Kahneman (1992).

ECONOMIC MECHANISM: A "categorical" perceptual distinction between certainty and near-certainty, or between impossibility and near-impossibility, could generate a very steep but technically continuous weighting function rather than a true jump discontinuity.

LITERATURE CONNECTION: Directly extends the endpoint-behavior discussion in this paper's own text and its resolution attempt in Tversky and Kahneman (1992).

DATA: Densely sampled choice data at probabilities within 0.1 percentage points of 0 and 100 percent, collected via a large incentivized online panel.

VARIABLES: Choice proportions at finely spaced probability values near the endpoints.

METHOD: Non-parametric curve estimation (e.g., local polynomial regression) of $\pi(p)$ recovered from certainty-equivalent elicitation near the endpoints.

IDENTIFICATION STRATEGY: Fine-grained probability spacing near the endpoints is the source of identifying variation not available in the original paper's coarser problem set (which jumps from $p=.001$ to $p=.20$ to $p=1$).

EXPECTED CONTRIBUTION: A precise, modern answer to a specific functional-form question the original paper raises but explicitly leaves open.

MAIN RISK: Extremely small and extremely large probabilities are notoriously difficult for respondents to process meaningfully at all, which could contaminate estimates near the very endpoints the design is meant to characterize.

12. Research Idea Ranking

Ranked by novelty, feasibility, data availability, methodological quality, economic importance, and potential contribution, the three strongest ideas above are, in order: Idea 2 (unrealized-loss trading behavior), Idea 1 (incentivized high-stakes replication), and Idea 4 (algorithmic delegation).

Idea 2 ranks first because brokerage panel data with cost-basis information already exists at scale, the identification strategy (within-account, within-time comparison across simultaneously held positions) is credible without requiring new data collection, and the question bears directly on live debates in the disposition-effect literature. Idea 1 ranks second: it is the most direct test of the paper's most frequently voiced criticism, is straightforward to implement with modern incentivized-experiment infrastructure, but is less novel because incentivized replications of

Allais-type problems already exist in some form. Idea 4 ranks third: it is highly novel, since algorithmic delegation is a genuinely new institutional setting, but selection into delegated accounts is a serious identification threat that would require either a natural experiment (a platform's staggered rollout of a robo-advisory product) or strong instruments to overcome, which lowers its near-term feasibility relative to Ideas 1 and 2.

13. Learning Framework

13.1 Prerequisite Concepts

Expected utility theory and its three tenets (expectation, asset integration, risk aversion via concave u); the von Neumann–Morgenstern axioms (completeness, transitivity, continuity, independence); basic conjoint measurement theory sufficient to follow the Appendix's additive and multiplicative representation argument; and the Allais paradox as a worked numerical example.

13.2 Core Concepts Learned

The certainty effect, reflection effect, and isolation effect as three empirically distinguishable signatures of expected-utility violation; the distinction between a value function defined on gains and losses versus a utility function defined on final wealth; the distinction between overweighting (a property of decision weights) and overestimation (a property of probability judgment); and the four defining properties of the decision-weighting function — overweighting of small probabilities, subadditivity, subcertainty, and subproportionality.

13.3 Important Equations

The bilinear evaluation rule for regular prospects, $V(x,p;y,q)=\pi(p)v(x)+\pi(q)v(y)$ (equation 1); the segregated evaluation rule for strictly positive or strictly negative prospects, $V(x,p;y,q)=v(y)+\pi(p)[v(x)-v(y)]$ (equation 2); the loss-aversion inequality $v(x)<-v(-x)$ and $v'(x)<v'(-x)$ (equation 4); and the subcertainty inequality $\pi(p)+\pi(1-p)<1$.

13.4 Methods Learned

The matched-pair, minimally transformed choice-problem design as a way of isolating exactly which axiom of a theory fails; deriving functional-form restrictions on unobserved value and weighting functions purely from ordinal choice reversals, without needing a cardinal utility elicitation procedure; and the additive-and-multiplicative conjoint-measurement route to a bilinear representation theorem.

13.5 Questions to Test Understanding

Comprehension. Can you state, in your own words, why the reflection effect rules out variance aversion as an alternative explanation of the certainty effect? Can you reproduce the algebraic step that turns the Problem 1/2 reversal into the subcertainty inequality $\pi(.66) + \pi(.34) < 1$? **Analysis.** Why does the paper need two separate evaluation equations (1 and 2) rather than one, and under what condition would equation (2) collapse into equation (1)? Why is overweighting of small probabilities alone insufficient to explain the observed rejection of probabilistic insurance, and what additional property of π is required? **Research.** Design a single choice problem that would yield different predictions from prospect theory, regret theory, and rank-dependent expected utility, and specify exactly which observed choice would favor which theory. If you found that the certainty effect vanished entirely under sufficiently large real incentives, would this refute prospect theory, or only its magnitude under the specific hypothetical-choice elicitation used here?

13.6 Papers to Read Next

Tversky and Kahneman (1992), for the parametric, cumulative extension that resolves several of this paper's own open questions; Rabin (2000), for the formal proof that expected utility cannot be rescued by any reasonable degree of risk aversion; and Barberis, Huang, and Santos (2001), for the market-level asset-pricing bridge this paper does not itself attempt.

14. Conclusion

14.1 One-Sentence Thesis

Choice under risk is governed not by the expectation of a concave utility function defined over final wealth, but by a reference-dependent value function, concave for gains and convex and steeper for losses, combined with a decision-weighting function that overweights small probabilities and underweights large ones relative to their stated values.

14.2 Three Most Important Insights

First, certainty is overweighted specifically because it is certain, not merely because it has a high probability — a distinct psychological category from "very likely." Second, the sign of an outcome relative to a reference point, not only its magnitude, is a first-order determinant of risk attitude, which no single monotone utility-of-wealth function can capture. Third, how a choice is described, not only its reduced-form probabilities and payoffs, can change which choice is made, because decision makers edit prospects — combining, segregating, and cancelling components — before they evaluate them.

14.3 Strongest Evidence

The formal proof, in Section 3.4 above, that probabilistic insurance is strictly preferred to full insurance under any concave expected-utility function, combined with the 80 percent majority that rejects probabilistic insurance in Problem 9, is the paper's most decisive single result: it is an exact mathematical incompatibility, not a matter of degree or estimation, between a specific real preference pattern and the entire class of concave-utility expected-utility models.

14.4 Biggest Weakness

The absence of any market-level implication or bridge: the paper is, by design, silent on whether the individual-level anomalies it documents survive aggregation, arbitrage, and learning well enough to shape asset prices, insurance markets, or gambling markets, leaving that entire question to three subsequent decades of behavioral finance research.

14.5 Most Important Research Gap

A general, not merely existence-proof, account of how much individual-level loss aversion and probability weighting survives in equilibrium once heterogeneous investors, arbitrage, and market aggregation are taken into account.

14.6 Best Research Opportunity

Testing the reference-point-adaptation prediction of Section 3.9 against real brokerage account data (Research Idea 2, Section 11), because the required data already exist at scale and the within-account identification strategy is credible without new data collection.

14.7 Three Papers to Read Next

Tversky and Kahneman (1992); Rabin (2000); Barberis, Huang, and Santos (2001).

15. References

- Allais, M. (1953). Le comportement de l'homme rationnel devant le risque: critique des postulats et axiomes de l'école américaine. *Econometrica*, 21(4), 503–546.
- Barberis, N., Huang, M., and Santos, T. (2001). Prospect theory and asset prices. *Quarterly Journal of Economics*, 116(1), 1–53.
- Benartzi, S., and Thaler, R. H. (1995). Myopic loss aversion and the equity premium puzzle. *Quarterly Journal of Economics*, 110(1), 73–92.
- De Long, J. B., Shleifer, A., Summers, L. H., and Waldmann, R. J. (1990). Noise trader risk in financial markets. *Journal of Political Economy*, 98(4), 703–738.
- Ellsberg, D. (1961). Risk, ambiguity, and the Savage axioms. *Quarterly Journal of Economics*, 75(4), 643–669.

- Edwards, W. (1962). Subjective probabilities inferred from decisions. *Psychological Review*, 69(2), 109–135.
- Fellner, W. (1961). Distortion of subjective probabilities as a reaction to uncertainty. *Quarterly Journal of Economics*, 75(4), 670–690.
- Fishburn, P. C., and Kochenberger, G. A. (1979). Two-piece von Neumann–Morgenstern utility functions. *Decision Sciences*, 10(4), 503–518.
- Friedman, M., and Savage, L. J. (1948). The utility analysis of choices involving risk. *Journal of Political Economy*, 56(4), 279–304.
- Grether, D. M., and Plott, C. R. (1979). Economic theory of choice and the preference reversal phenomenon. *American Economic Review*, 69(4), 623–638.
- Kahneman, D., and Tversky, A. (1979). Prospect theory: An analysis of decision under risk. *Econometrica*, 47(2), 263–291.
- Lichtenstein, S., and Slovic, P. (1971). Reversal of preferences between bids and choices in gambling decisions. *Journal of Experimental Psychology*, 89(1), 46–55.
- Loomes, G., and Sugden, R. (1982). Regret theory: An alternative theory of rational choice under uncertainty. *Economic Journal*, 92(368), 805–824.
- Markowitz, H. (1952). The utility of wealth. *Journal of Political Economy*, 60(2), 151–158.
- Odean, T. (1998). Are investors reluctant to realize their losses? *Journal of Finance*, 53(5), 1775–1798.
- Quiggin, J. (1982). A theory of anticipated utility. *Journal of Economic Behavior and Organization*, 3(4), 323–343.
- Rabin, M. (2000). Risk aversion and expected-utility theory: A calibration theorem. *Econometrica*, 68(5), 1281–1292.
- Shefrin, H., and Statman, M. (1985). The disposition to sell winners too early and ride losers too long: Theory and evidence. *Journal of Finance*, 40(3), 777–790.
- Tversky, A., and Kahneman, D. (1974). Judgment under uncertainty: Heuristics and biases. *Science*, 185(4157), 1124–1131.
- Tversky, A., and Kahneman, D. (1992). Advances in prospect theory: Cumulative representation of uncertainty. *Journal of Risk and Uncertainty*, 5(4), 297–323.